

ESARAPU VARUN

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SUMMARY

Electronics and Communication Engineering graduate with experience in Python, machine learning, and backend development using Django REST Framework. Built end-to-end applications involving API design, data processing, and system workflows. Skilled in TensorFlow, scikit-learn, Pandas, and SQL with strong problem-solving and communication abilities.

WORK EXPERIENCE

AI Operator Intern

XTRANSMATRIX CONSULTING SERVICES PVT LTD | Nov 2025 - Mar 2026

- Collaborated with the AI/ML team to develop, test, and optimize machine learning models, bridging the gap between raw data and actionable AI insights.
 - Performed Human-in-the-Loop (HITL) data annotation, cleaning, and preprocessing using Python, NumPy, and Pandas to ensure high-quality training and evaluation datasets.
 - Designed and implemented supervised and unsupervised learning models, including predictive modeling and clustering, to support end-to-end project workflows.
 - Managed sensitive data with adherence to confidentiality and security standards while actively documenting project outcomes and emerging AI solutions.
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SKILLS SUMMARY

Languages : Python,C, SQL.
Databases : MySQL, MongoDB.
Backend : Django , Django REST Framework
Machine Learning / Data Science : Pandas, NumPy, scikit-learn, Matplotlib, TensorFlow, Keras, PyTorch.
Tools : Git, Jupyter Notebook, Visual Studio Code, IntelliJ IDEA.
Cloud : AWS (basic : EC2 , S3)
Soft Skills : Rapport Building, People Management, Public Speaking.
Concepts : Data Preprocessing, Deep Learning, NLP, CNN, LSTM, Recommendation Systems

EDUCATION

HKBK College Of Engineering | Bangalore, KA | Dec 2022 - Jul 2026

B.E. in Electronics and Communication Engineering

GPA:8.71/10

PROJECTS

Vehicle Service Management System | [LINK](#) | Apr 2026

Built **Backend RESTful APIs** using **Django REST Framework** to manage vehicles, components, issues, and payments. Implemented **issue lifecycle management** with **status transitions** and **validation logic**. Designed **cost calculation** combining labor and component pricing. Developed a **Streamlit interface** for updating status, simulating payments, and visualizing revenue.

Real-time Speech Emotion Recognition system | [LINK](#) | Oct 2025

Extracted **MFCC** features using **Librosa**. Applied **preprocessing** including **normalization** and **padding**. Designed and trained a **CNN-based machine learning model** for classification. Performed **model evaluation** and **optimization** to improve performance and accuracy

Text Emotion Classification Model using LSTM | [LINK](#) | Aug 2025

Built an **LSTM-based NLP model for emotion classification with 87% accuracy** using TensorFlow/Keras. Handled text preprocessing (tokenization, padding, normalization) on custom datasets. Utilized Python libraries including **Pandas**, **NumPy**, **scikit-learn**, **TensorFlow**, and **Keras** for model building and evaluation

Personalized Movie Recommendation System | [LINK](#) | Apr 2025

Engineered a **neural collaborative filtering model** using MovieLens dataset, **attaining 89.2% prediction accuracy**. Developed and optimized a machine learning model for classification tasks using **Python** and **TensorFlow**.

CERTIFICATES

Python Programming (SoloLearn) | [CERTIFICATE](#) | Apr 2022

Java (SoloLearn) | [CERTIFICATE](#) | Nov 2022

Scientific Computing with Python (FreeCodeCamp) | [CERTIFICATE](#) | Jul 2022

Machine Learning with Python (FreeCodeCamp) | [CERTIFICATE](#) | Aug 2022

Architecting SAP on AWS (Amazon Web Services) | [CERTIFICATE](#) | Nov 2025

EXTRA-CURRICULAR ACTIVITIES

- Hosted and coordinated multiple **college fests**, managing logistics, sponsors, and student teams for smooth execution
- Organized domain-specific **ECE workshops (VACs)** with 60+ participants, ensuring effective content delivery and engagement
- Served as **Event Coordinator** for Technisium 2025, overseeing 8 technical events with **150+ participating teams** from all over Karnataka.